

Technical Document #563: Database Systems - Comprehensive Overview

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Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Database indexing improves query performance by creating data structures that allow faster data retrieval.

Database monitoring tracks performance metrics and identifies bottlenecks. Query analysis and indexing strategies optimize database performance.

Database caching stores frequently accessed data in memory. Redis and Memcached are popular caching solutions.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Database connection pooling reuses database connections to reduce overhead.
- ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.
- Database migration tools manage schema changes and data transformations.